Customer Shopping Behavior Analysis

An End-to-End Data Analytics Project

Data Cleaning (Python) · SQL Analysis (MySQL) · Dashboard (Power BI)

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Dataset: 3,900 customer transaction records · 18 attributes

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Executive Summary

The **Customer Shopping Behavior Analysis** project is an end-to-end data analytics project that demonstrates the complete process of transforming raw retail transaction data into meaningful business insights. The primary objective of this project is to analyze customer purchasing behavior, identify key sales patterns, and provide data-driven recommendations that can support better business decision-making.

The project utilizes a retail customer shopping dataset containing information such as customer demographics, purchased products, product categories, purchase amounts, review ratings, subscription status, payment methods, shipping preferences, discounts, and previous purchase history. The dataset was first cleaned and preprocessed using **Python**, where missing values, duplicate records, and data inconsistencies were identified and addressed. Exploratory Data Analysis (EDA) was then performed to understand data distributions, customer behavior, and relationships between important variables through statistical summaries and visualizations.

After data preparation, the cleaned dataset was analyzed using **MySQL** to answer business-oriented questions related to customer segmentation, product performance, purchase trends, subscription behavior, discount usage, and sales analysis. These SQL queries helped extract meaningful insights that support strategic business decisions.

Finally, an interactive **Power BI dashboard** was developed to visualize key performance indicators (KPIs), customer demographics, product category performance, purchase behavior, and other important business metrics. The dashboard enables users to explore the data interactively through filters and visual reports, making complex information easier to understand.

The analysis revealed valuable insights into customer purchasing patterns, high-performing product categories, and customer engagement. Based on these findings, practical business recommendations were proposed, including improving customer retention strategies, optimizing discount campaigns, enhancing subscription benefits, and focusing marketing efforts on high-value customer segments.

Overall, this project demonstrates practical skills in **Python, SQL, Power BI, data cleaning, exploratory data analysis, business intelligence, and data visualization**. It reflects a complete real-world data analytics workflow and showcases the ability to convert raw data into actionable business insights that support informed decision-making.

Project Overview & Objectives

1. Project Overview :

- **Customer Shopping Behavior Analysis :**

The **Customer Shopping Behavior Analysis** project is an end-to-end data analytics project designed to analyze retail customer purchasing behavior using modern data analytics tools and techniques. The project follows a complete analytics workflow, beginning with data collection and preprocessing, followed by exploratory data analysis, SQL-based business analysis, interactive dashboard development, and business reporting.

The dataset contains customer demographic information, purchase history, product categories, payment methods, shipping preferences, review ratings, subscription status, discounts, and other transactional details. By analyzing these data points, the project aims to identify meaningful patterns in customer behavior, understand purchasing trends, and extract insights that can support business decision-making.

Python was used for data cleaning, preprocessing, and exploratory data analysis (EDA). SQL was employed to answer business-focused questions and generate analytical insights from the cleaned dataset. Finally, Power BI was used to develop an interactive dashboard that presents key performance indicators (KPIs), customer demographics, product performance, and purchasing trends in an intuitive and visually appealing format.

This project demonstrates the practical application of data analytics techniques to solve real-world business problems and showcases the complete process of converting raw data into actionable business intelligence.

2. Project Objectives

The primary objectives of this project are:

- To perform data cleaning and preprocessing to improve data quality and prepare the dataset for analysis.
- To conduct Exploratory Data Analysis (EDA) in order to understand customer demographics, purchasing behavior, and product trends.
- To identify patterns, relationships, and anomalies within the customer shopping data through statistical analysis and data visualization.
- To write SQL queries that answer important business questions related to customer behavior, sales performance, product categories, and purchasing trends.
- To develop an interactive Power BI dashboard that effectively visualizes key business metrics and supports data-driven decision-making.

- To generate meaningful business insights and provide practical recommendations that can help improve customer engagement, marketing strategies, and overall business performance.
- To demonstrate proficiency in Python, SQL, Power BI, data cleaning, data visualization, and business analytics through a complete end-to-end analytics workflow.

Dataset Overview

The **Customer Shopping Behavior** dataset is a retail transaction dataset containing information about customer demographics, purchasing patterns, product categories, payment methods, shipping preferences, discounts, and subscription status. It is used to analyze customer behavior, identify sales trends, and generate business insights.

The dataset consists of **3,900 records** and **18 features**, providing sufficient information for data cleaning, exploratory data analysis (EDA), SQL-based business analysis, and Power BI dashboard development.

Dataset Details :

Attribute	Details
Dataset Name	Customer Shopping Behavior
Source	Kaggle
Domain	Retail / E-Commerce
File Format	CSV
Records	3,900
Features	18

The dataset includes key attributes such as **Age, Gender, Item Purchased, Category, Purchase Amount, Review Rating, Subscription Status, Shipping Type, Payment Method, Previous Purchases,** and **Purchase Frequency**, making it suitable for analyzing customer shopping trends and supporting data-driven business decisions.

```
[4]: df.head()
```

[4]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previc Purchas
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	

Key Findings by Business Question

Business Question	Key Finding
1. Which product category generated the highest revenue?	Clothing contributed the highest overall revenue among all product categories.
2. Which age group spends the most?	Young Adults (26–35 years) recorded the highest total purchase amount.
3. Does subscription status affect purchasing behavior?	Subscription status showed only a small difference in average purchase amount between subscribed and non-subscribed customers.
4. Which payment methods are most commonly used?	Credit Card and PayPal were among the most frequently used payment methods.
5. Which shipping method is preferred?	Standard Shipping was the most commonly selected shipping option.
6. Do discounts influence customer purchases?	Products with discounts attracted more purchases, indicating that promotional offers positively influenced buying decisions.
7. Which products receive the highest customer ratings?	Several clothing and footwear products consistently received high review ratings.
8. Which season records the highest sales?	Sales remained relatively balanced across seasons, with slight variations depending on product categories.
9. Who are the most loyal customers?	Customers with a higher number of previous purchases demonstrated stronger loyalty and repeat purchasing behavior.
10. What business insights can improve sales?	Focusing on high-performing product categories, optimizing promotional campaigns, and strengthening customer loyalty programs can improve overall business performance.

```sql
-- q1. what is the total revenue generated by male vs. female customers?
select gender,
       sum(purchase_amount) as revenue
from customer
group by gender;

-- q2. which customers used a discount but still spent more than the average purchase amount?
select customer_id,
       purchase_amount
from customer
where discount_applied = 'Yes'
  and purchase_amount >= (
      select avg(purchase_amount)
      from customer
  );

-- q3. which are the top 5 products with the highest average review rating?
select item_purchased,
       round(avg(review_rating), 2) as average_product_rating
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;

-- q4. compare the average purchase amounts between standard and express shipping.
select shipping_type,
       round(avg(purchase_amount), 2) as average_purchase
from customer
where shipping_type in ('Standard', 'Express')
group by shipping_type;
```

```sql
-- q5. do subscribed customers spend more? compare average spend and total revenue between subscribers and non-subscribers.
select subscription_status,
       count(customer_id) as total_customers,
       round(avg(purchase_amount), 2) as avg_spend,
       round(sum(purchase_amount), 2) as total_revenue
from customer
group by subscription_status
order by total_revenue desc, avg_spend desc;

-- q6. which 5 products have the highest percentage of purchases with discounts applied?
select item_purchased,
       round(
          100 * sum(case
                     when discount_applied = 'Yes' then 1
                     else 0
                   end) / count(*), 2
       ) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

```sql
-- q7. segment customers into new, returning, and loyal based on their previous purchases.
select
    case
        when previous_purchases <= 1 then 'New'
        when previous_purchases between 2 and 10 then 'Returning'
        else 'Loyal'
    end as customer_segment,
    count(*) as number_of_customers
from customer
group by customer_segment;
```

```sql
-- q8. what are the top 3 most purchased products within each category?
with item_counts as (
    select
        category,
        item_purchased,
        count(customer_id) as total_orders,
        row_number() over (
            partition by category
            order by count(customer_id) desc
        ) as item_rank
    from customer
    group by category, item_purchased
)
select item_rank,
       category,
       item_purchased,
       total_orders
from item_counts
where item_rank <= 3
order by category, item_rank;

-- q9. are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
select subscription_status,
       count(customer_id) as repeat_buyers
from customer
where previous_purchases > 5
group by subscription_status;
```

```sql
-- q10. what is the revenue contribution of each age group?
select age_group,
SUM(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc;
```

Additional Insights

Beyond the primary business questions, the analysis revealed several additional observations that provide a deeper understanding of customer shopping behavior:

- Customers across different age groups showed relatively consistent purchasing activity, indicating a broad customer base.
- Clothing and Accessories were among the most frequently purchased product categories, contributing significantly to overall sales.
- Standard Shipping was the most preferred shipping method, suggesting customers prioritize cost-effective delivery options.
- Credit Card, PayPal, and Cash were the most commonly used payment methods, reflecting diverse customer payment preferences.
- Customers with a higher number of previous purchases demonstrated stronger loyalty and were more likely to make repeat purchases.
- Discounts and promotional offers positively influenced purchasing behavior, encouraging higher customer engagement.
- Product review ratings remained generally high, indicating overall customer satisfaction with purchased products.
- The interactive Power BI dashboard enabled effective monitoring of key business metrics and simplified the analysis of customer behavior through dynamic visualizations.

Power BI Dashboard

Recommendations

Based on the analysis of customer shopping behavior, the following recommendations are proposed to improve business performance and customer satisfaction:

1. **Focus on High-Performing Product Categories**
   Increase inventory and promotional efforts for product categories that consistently generate higher sales and customer demand.
2. **Strengthen Customer Loyalty Programs**
   Reward repeat customers with exclusive offers, discounts, or loyalty points to encourage long-term customer retention.
3. **Optimize Promotional Campaigns**
   Use discounts and promotional codes strategically to increase sales while maintaining healthy profit margins.
4. **Enhance Subscription Benefits**
   Introduce additional benefits for subscribed customers, such as free shipping or exclusive discounts, to improve subscription adoption and retention.
5. **Improve Marketing Strategies**
   Develop targeted marketing campaigns based on customer demographics, purchase history, and shopping preferences to increase engagement and conversions.
6. **Monitor Customer Feedback**
   Regularly analyze customer review ratings and feedback to identify improvement areas and enhance product quality and customer experience.
7. **Leverage Interactive Dashboards**
   Continue using Power BI dashboards to monitor key performance indicators (KPIs), track customer behavior, and support data-driven decision-making.

Conclusion

The **Customer Shopping Behavior Analysis** project successfully demonstrated an end-to-end data analytics workflow using **Python, MySQL, and Power BI**. The dataset was cleaned, explored, and analyzed to understand customer purchasing behavior, product performance, and sales trends.

Through exploratory data analysis, SQL-based business queries, and interactive dashboard development, the project identified valuable insights into customer demographics, purchasing patterns, payment preferences, and product categories. These findings were used to provide practical recommendations that can support better business decisions, improve customer engagement, and optimize sales strategies.

Overall, this project highlights the importance of data analytics in transforming raw data into actionable business intelligence. It also demonstrates practical skills in **data cleaning, exploratory data analysis (EDA), SQL querying, data visualization, dashboard development, and business reporting**, making it a strong example of an end-to-end data analytics project.